Dokumentasi Konfigurasi Lab 4

1. Pemeriksaan sebelum konfigurasi

Langkah 1 - Buat backup baru

Pada terminal winbox gunakan command : /system backup save name=sebelum-lab4

```
  MMM       MMM       KKK                          TTTTTTTTTTT      KKK
  MMMM     MMMM       KKK                          TTTTTTTTTTT      KKK
  MMM MMMM MMM  III   KKK  KKK  RRRRRR     OOOOOO      TTT     III  KKK  KKK
  MMM  MM  MMM  III   KKKKK     RRR  RRR  OOO  OOO     TTT     III  KKKKK
  MMM       MMM  III   KKK KKK   RRRRRR    OOO  OOO     TTT     III  KKK KKK
  MMM       MMM  III   KKK  KKK  RRR  RRR   OOOOOO      TTT     III  KKK  KKK

  MikroTik RouterOS 6.39.2 (c) 1999-2017       http://www.mikrotik.com/

[?]             Gives the list of available commands
command [?]     Gives help on the command and list of arguments

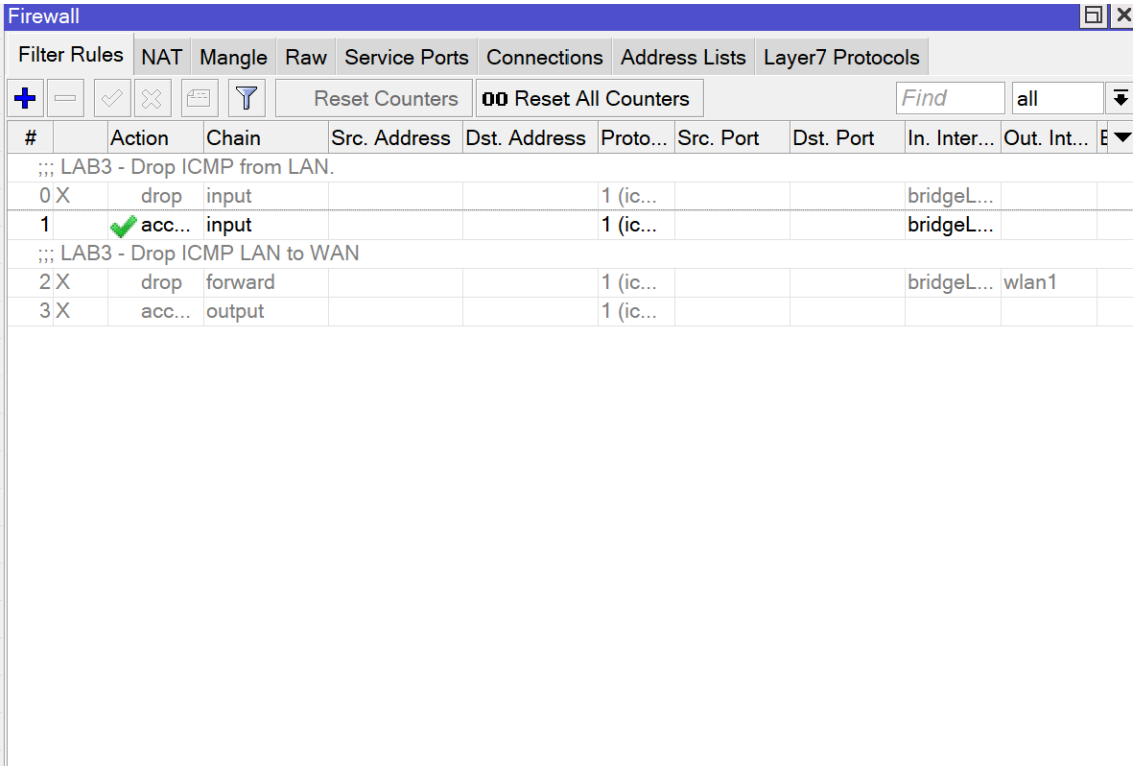
[Tab]           Completes the command/word. If the input is ambiguous,
                a second [Tab] gives possible options

/               Move up to base level
..              Move up one level
/command        Use command at the base level
[admin@MikroTik] > /system backup save name=sebelum-lab4
Saving system configuration
Configuration backup saved
[admin@MikroTik] >
```

Gambar 1. Melakukan backup sistem sebelum konfigurasi

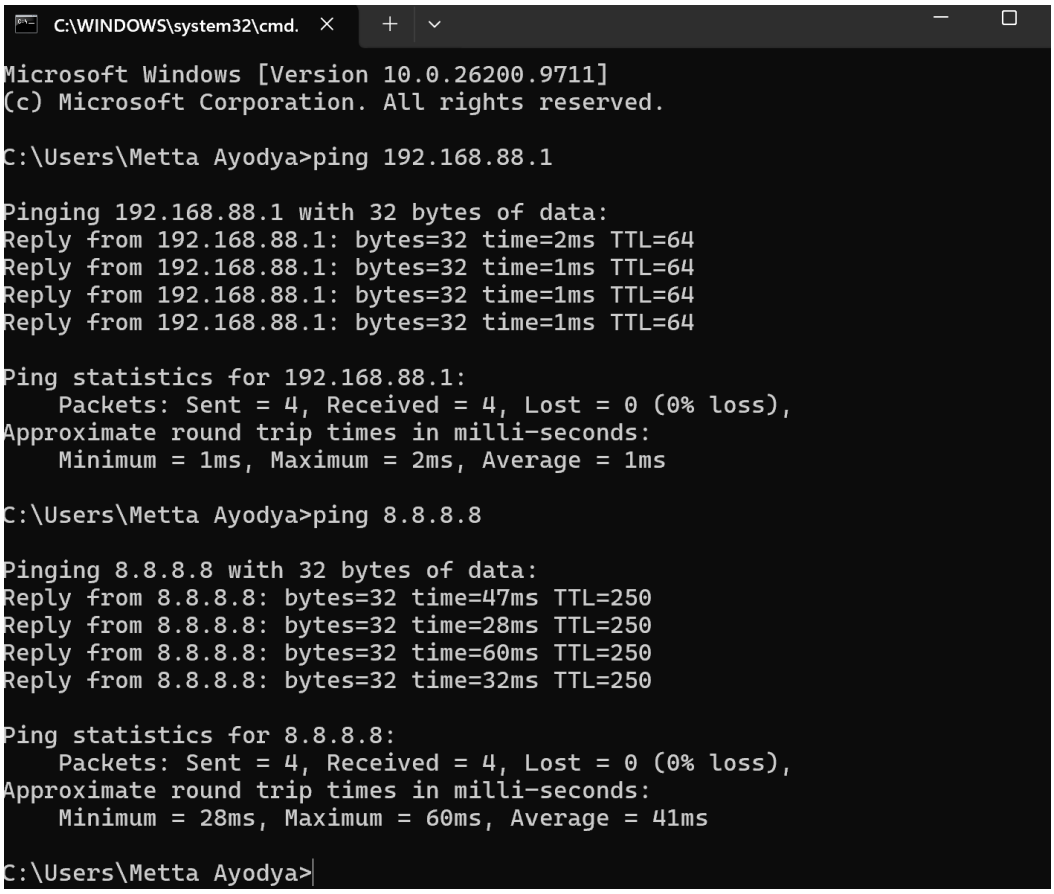
Langkah 2 - Periksa Filter Rules

Pastikan semua filter rules drop sudah di nonaktifkan



Gambar 2. Memastikan filter rule yang tidak digunakan dinonaktifkan

Langkah 3 - Memastikan konektivitas normal

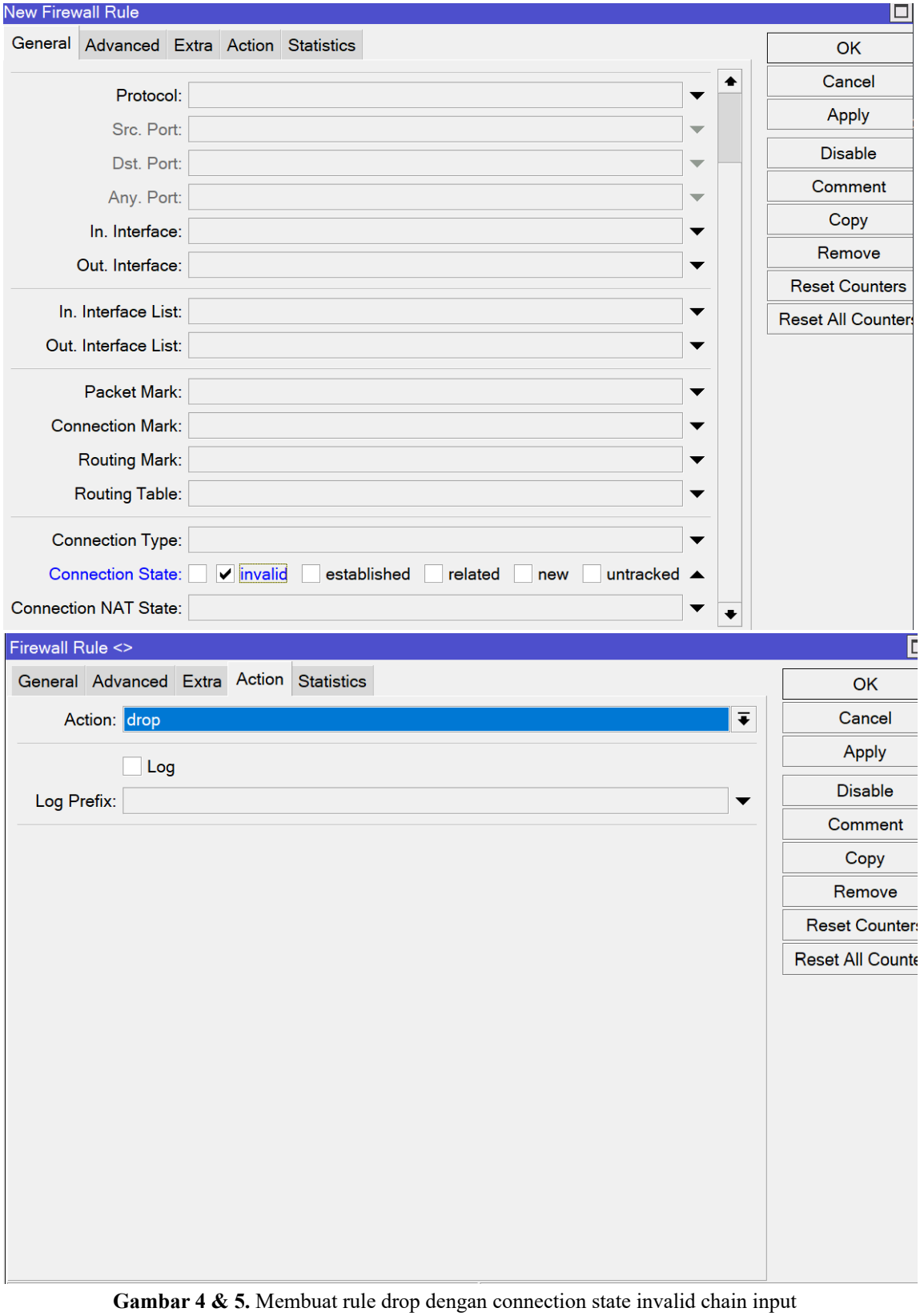


Gambar 3. Memastikan konektivitas aman setelah konfigurasi dilakukan

2. Membuat Connection State Drop (Mengarah ke Input)

Langkah 1 - tambahkan rule drop invalid

Bisa melalui GUI langsung atau via terminal menggunakan command berikut /ip firewall filter add chain=input connection-state=invalid action=drop comment="LAB4 - Drop Invalid Input"



Gambar 4 & 5. Membuat rule drop dengan connection state invalid chain input

Langkah 3 – Cek Konektivitas

```
Ping statistics for 192.168.88.1:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 1ms, Maximum = 2ms, Average = 1ms

C:\Users\Metta Ayodya>ping 8.8.8.8

Pinging 8.8.8.8 with 32 bytes of data:
Reply from 8.8.8.8: bytes=32 time=50ms TTL=250
Reply from 8.8.8.8: bytes=32 time=27ms TTL=250
Reply from 8.8.8.8: bytes=32 time=28ms TTL=250
Reply from 8.8.8.8: bytes=32 time=36ms TTL=250

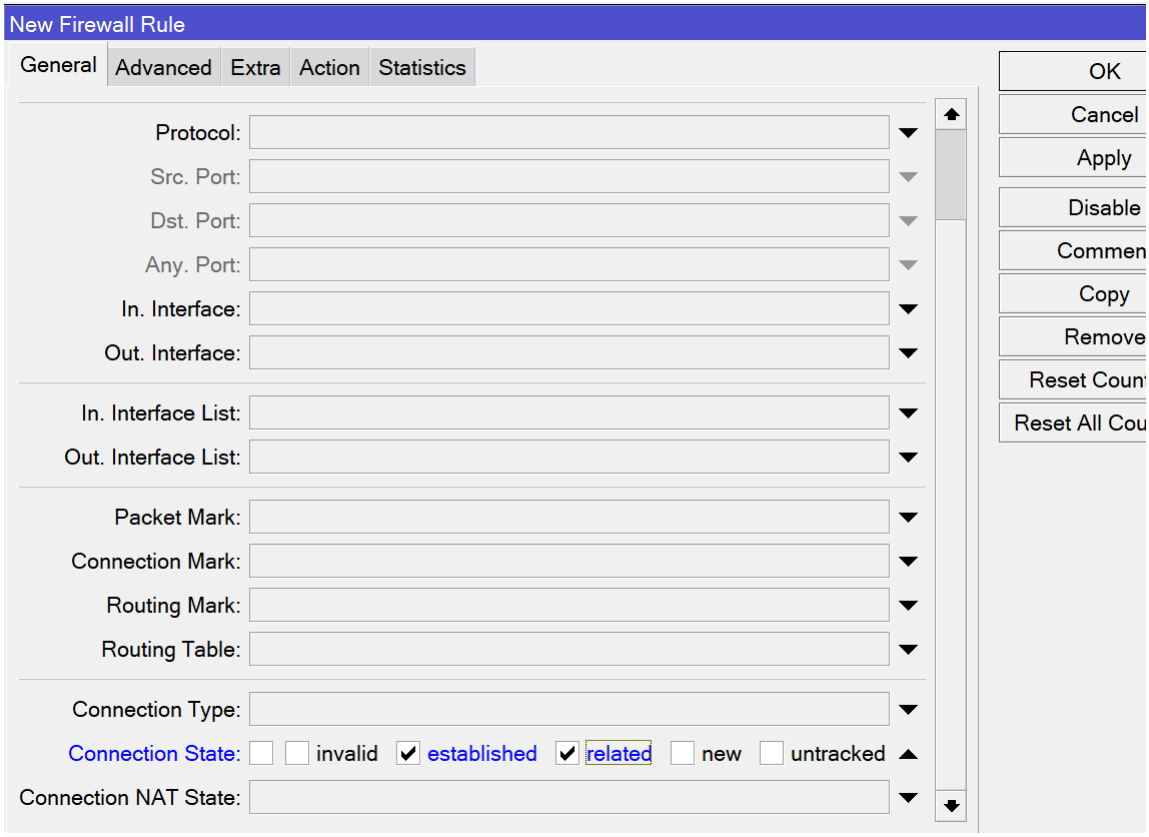
Ping statistics for 8.8.8.8:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 27ms, Maximum = 50ms, Average = 35ms

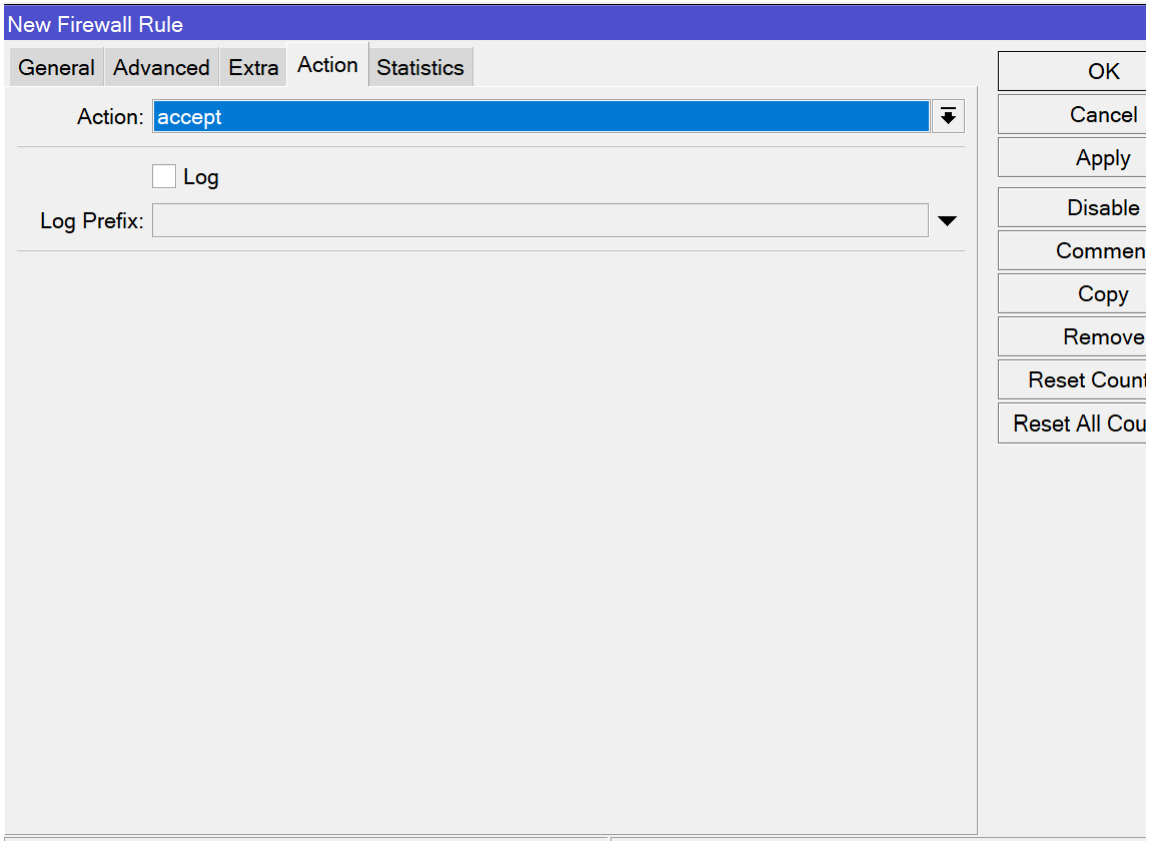
C:\Users\Metta Ayodya>
```

Gambar 6. Memastikan konektivitas aman setelah menambah rule

3. Menambahkan Connection State Established dan Related (Mengarah ke Input)

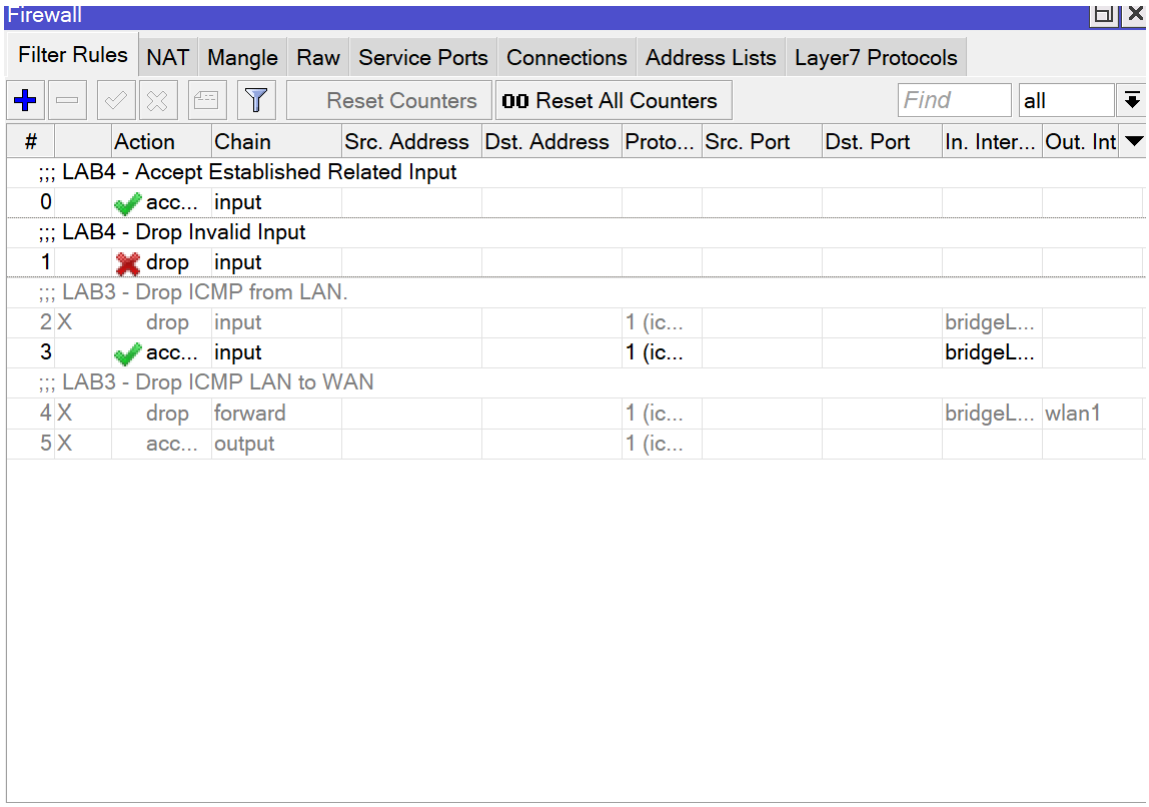
Langkah 1 - menambahkan rule accept state established dan related





Gambar 7. Menambah rule accept dengan connection state established dan related chain input

Langkah 2 - Pindahkan rule yang baru dibuat ke paling atas



Gambar 8. Memastikan urutan rule sesuai

Langkah 3 – cek konektivitas

```
C:\Users\Metta Ayodya>ping 192.168.88.1

Pinging 192.168.88.1 with 32 bytes of data:
Reply from 192.168.88.1: bytes=32 time=2ms TTL=64
Reply from 192.168.88.1: bytes=32 time=2ms TTL=64
Reply from 192.168.88.1: bytes=32 time=2ms TTL=64
Reply from 192.168.88.1: bytes=32 time=3ms TTL=64

Ping statistics for 192.168.88.1:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 2ms, Maximum = 3ms, Average = 2ms

C:\Users\Metta Ayodya>ping 8.8.8.8

Pinging 8.8.8.8 with 32 bytes of data:
Reply from 8.8.8.8: bytes=32 time=31ms TTL=250
Reply from 8.8.8.8: bytes=32 time=30ms TTL=250
Reply from 8.8.8.8: bytes=32 time=26ms TTL=250
Reply from 8.8.8.8: bytes=32 time=45ms TTL=250

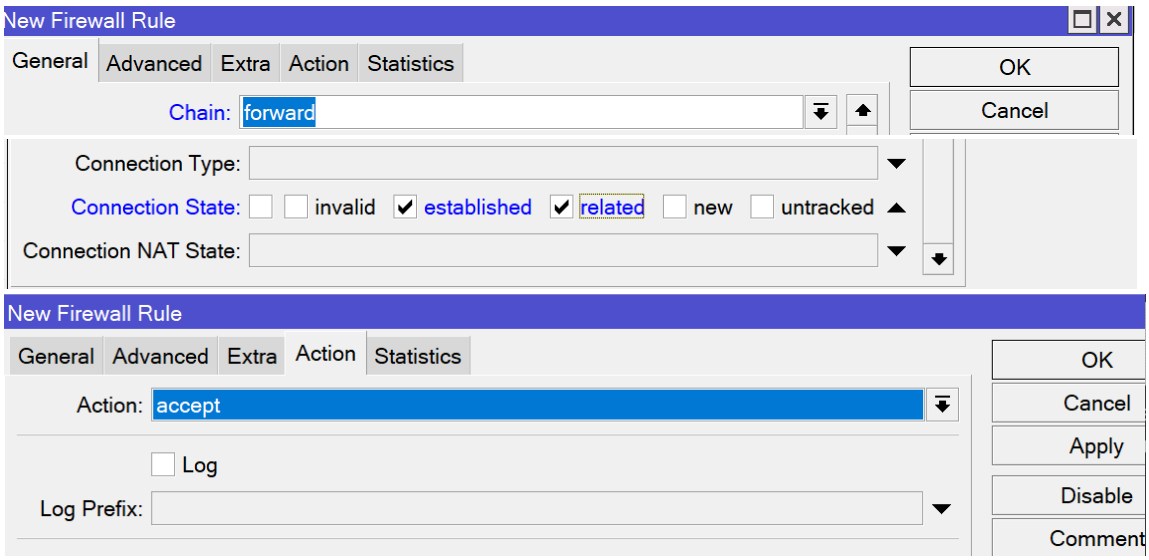
Ping statistics for 8.8.8.8:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 26ms, Maximum = 45ms, Average = 33ms

C:\Users\Metta Ayodya>
```

Gambar 9. Memastikan konektivitas tetap terhubung

4. Membuat rule accept established, related pada chain forward

Langkah 1 — Buat rule Accept Established, Related untuk forward



Gambar 10. Membuat rule accept chain forward dengan connection state established dan related

Langkah 2 – Cek Konektivitas

```
C:\Users\Metta Ayodya>ping 192.168.88.1

Pinging 192.168.88.1 with 32 bytes of data:
Reply from 192.168.88.1: bytes=32 time=2ms TTL=64
Reply from 192.168.88.1: bytes=32 time=2ms TTL=64
Reply from 192.168.88.1: bytes=32 time=1ms TTL=64
Reply from 192.168.88.1: bytes=32 time=2ms TTL=64

Ping statistics for 192.168.88.1:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 1ms, Maximum = 2ms, Average = 1ms

C:\Users\Metta Ayodya>ping 8.8.8.8

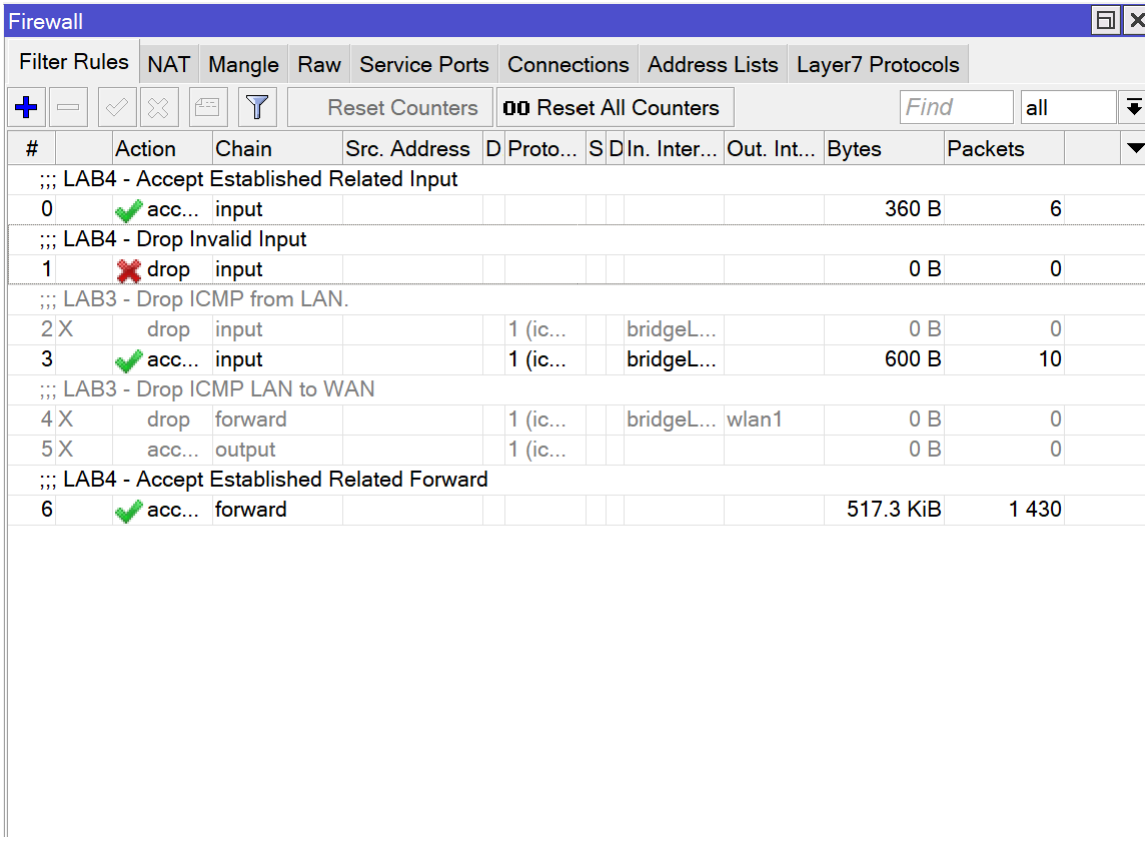
Pinging 8.8.8.8 with 32 bytes of data:
Reply from 8.8.8.8: bytes=32 time=36ms TTL=250
Reply from 8.8.8.8: bytes=32 time=35ms TTL=250
Reply from 8.8.8.8: bytes=32 time=20ms TTL=250
Reply from 8.8.8.8: bytes=32 time=145ms TTL=250

Ping statistics for 8.8.8.8:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 20ms, Maximum = 145ms, Average = 59ms

C:\Users\Metta Ayodya>
```

Gambar 11. Melakukan pengecekan konektivitas

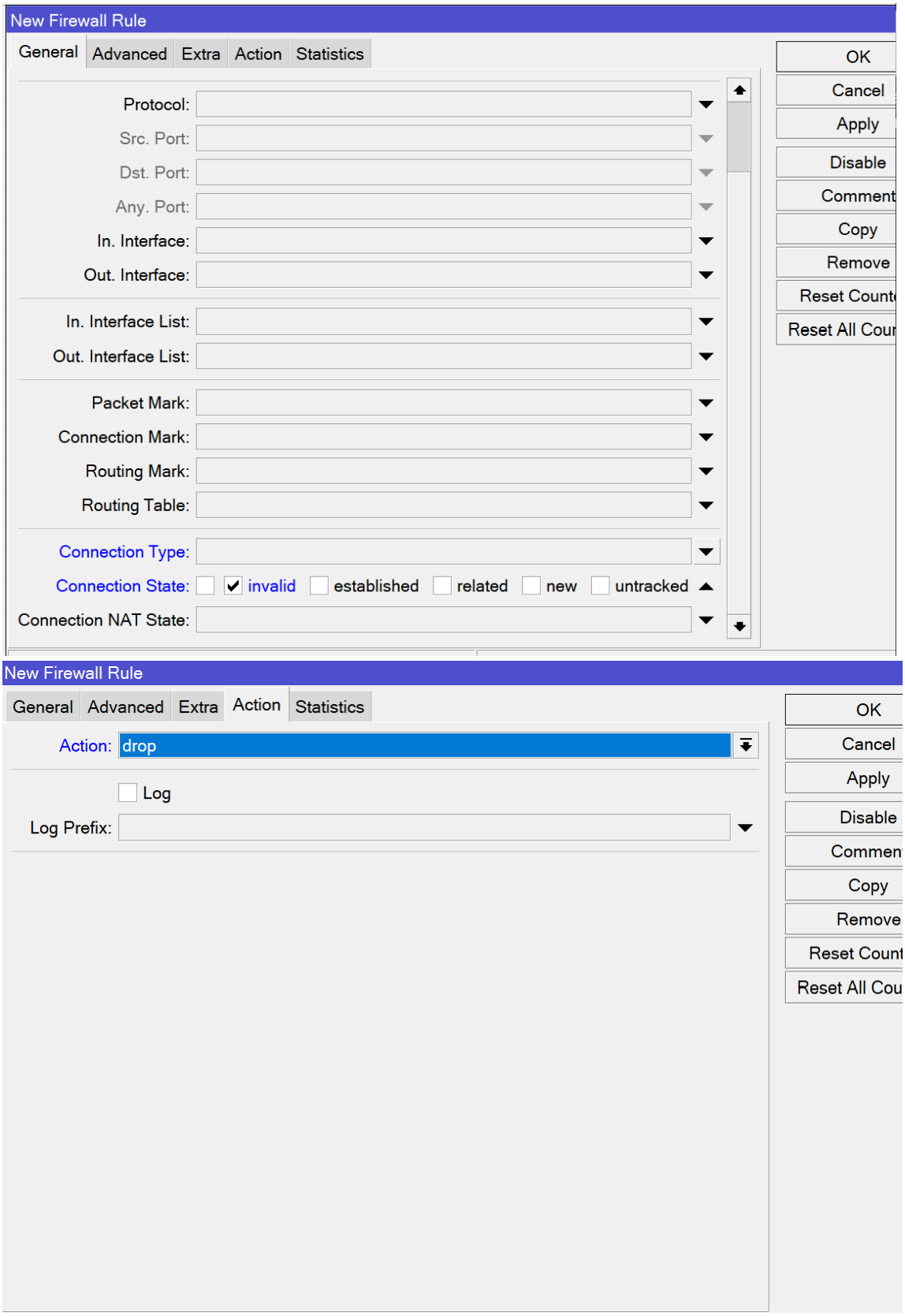
Langkah 4 – Cek Counter Packet



Gambar 12. Cek counter packet apakah ada perubahan setelah cek konektivitas

5. Tambahkan Rule Drop Invalid Chain Forward

Langkah 1 – Meanmbah rule drop baru pada forward chain dengan state invalid



Gambar 13 & 14. Menambahkan rule drop connection state invalid chain forward

Langkah 2 – Cek konektivitas

```
C:\Users\Metta Ayodya>ping 192.168.88.1

Pinging 192.168.88.1 with 32 bytes of data:
Reply from 192.168.88.1: bytes=32 time=2ms TTL=64
Reply from 192.168.88.1: bytes=32 time=2ms TTL=64
Reply from 192.168.88.1: bytes=32 time=1ms TTL=64
Reply from 192.168.88.1: bytes=32 time=2ms TTL=64

Ping statistics for 192.168.88.1:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 1ms, Maximum = 2ms, Average = 1ms

C:\Users\Metta Ayodya>ping 8.8.8.8

Pinging 8.8.8.8 with 32 bytes of data:
Reply from 8.8.8.8: bytes=32 time=36ms TTL=250
Reply from 8.8.8.8: bytes=32 time=35ms TTL=250
Reply from 8.8.8.8: bytes=32 time=20ms TTL=250
Reply from 8.8.8.8: bytes=32 time=145ms TTL=250

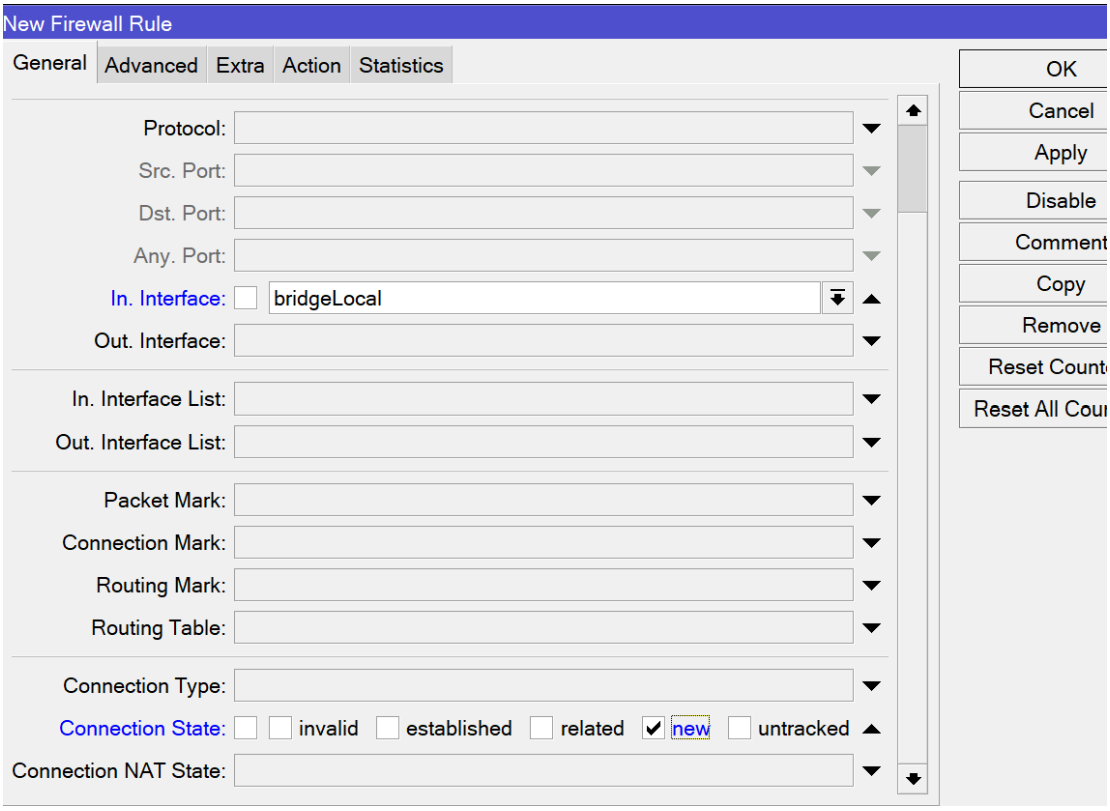
Ping statistics for 8.8.8.8:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 20ms, Maximum = 145ms, Average = 59ms

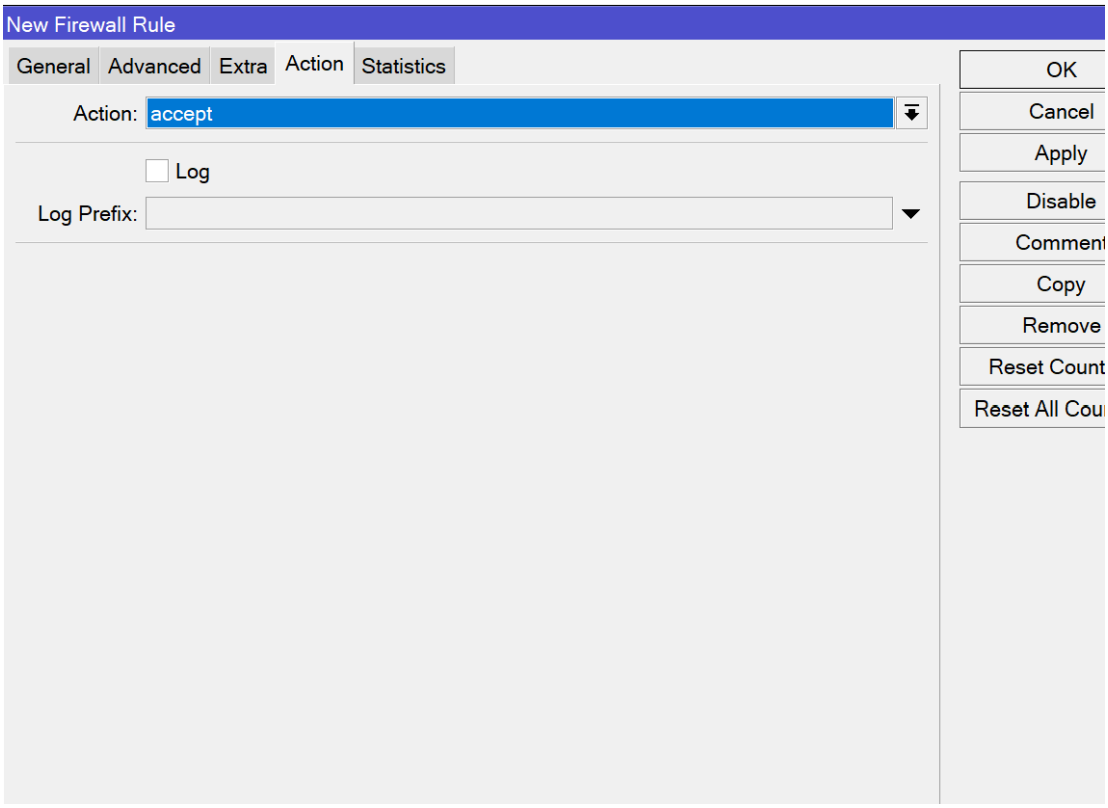
C:\Users\Metta Ayodya>
```

Gambar 16. Melakukan cek konektivitas dan hasilnya aman

6. Mengizinkan koneksi baru dari LAN

Langkah 1 – Menambahkan rule dengan connection state new





Gambar 17. Menambahkan rule accept baru dengan connection state new chain forward

Langkah 2 – Cek Konektivitas

```
C:\Users\Metta Ayodya>ping 192.168.88.1

Pinging 192.168.88.1 with 32 bytes of data:
Reply from 192.168.88.1: bytes=32 time=2ms TTL=64
Reply from 192.168.88.1: bytes=32 time=2ms TTL=64
Reply from 192.168.88.1: bytes=32 time=1ms TTL=64
Reply from 192.168.88.1: bytes=32 time=2ms TTL=64

Ping statistics for 192.168.88.1:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 1ms, Maximum = 2ms, Average = 1ms

C:\Users\Metta Ayodya>ping 8.8.8.8

Pinging 8.8.8.8 with 32 bytes of data:
Reply from 8.8.8.8: bytes=32 time=36ms TTL=250
Reply from 8.8.8.8: bytes=32 time=35ms TTL=250
Reply from 8.8.8.8: bytes=32 time=20ms TTL=250
Reply from 8.8.8.8: bytes=32 time=145ms TTL=250

Ping statistics for 8.8.8.8:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 20ms, Maximum = 145ms, Average = 59ms

C:\Users\Metta Ayodya>
```

Gambar 18. Melakukan cek konektivitas dengan hasil yang sesuai